PROGRAMMING TECHNIQUE II (SECJ1023)

Title : 13. CLIMATE ACTION

GROUP 7 (Climate Guardians) MEMBERS

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Table of Contents

No.	Contents	Pages
1.	Introduction	
2.	Problem Background and Proposed Solution	
3.	Objectives	
4.	Scopes	

1.0 Introduction

Climate change refers to long-term changes in average weather conditions that differentiate Earth's local, regional, and global climates . It is mostly caused by human activities such as the combustion of fossil fuels, deforestation, and industrial processes and is currently one of the most crucial and pressing problems for humanity. The accumulation of greenhouse gasses in the atmosphere led to global warming, which brought about large-scale and many possible irreversible changes in weather phenomena, sea levels, and ecosystems. All these changes present significant risks to food production, water availability, health, and lives, and the lifestyle of the populations, especially among the poorest people and developing countries.

Recognizing the gravity of the climate crisis and its far-reaching consequences for both the environment and society, the international community has established a global agenda to address climate change through Sustainable Development Goal 13 (SDG 13) "Climate Action": "Take urgent action to combat climate change and its impacts." Climate action refers to efforts and initiatives aimed at mitigating and adapting to the impacts of climate change. With the growing recognition of the severe and potentially irreversible consequences of climate change, there has been an increasing global focus on taking action to reduce greenhouse gas emissions, protect vulnerable communities, and transition to a more sustainable and resilient future.

Through Sustainable Development Goal 13 (SDG 13), this project focuses on climate action, which includes a variety of strategies and initiatives aimed at mitigating and adapting to the effects of climate change. The project will investigate key issues, challenges, and gaps in climate action before proposing innovative and effective solutions across sectors and levels.

2.0 Problem Background and Proposed Solution

Problem Statement

A:Clearly articulate the problem statement, highlighting its significance in the context of the selected SDG.

Climate change-induced rising temperatures and increased frequency of extreme weather events endanger human lives, infrastructure, and ecosystems. Heat waves due to the high temperatures, the meltdown of polar ice caps that causes the sea level to rise, severe storms, flood, and drought are all attributable to the high temperature . These challenges highlight the critical need to develop and implement sustainable methodologies to reduce carbon footprints, counter adverse impacts, and foster a carbon-free economy.

B:Discuss how addressing this problem can contribute to the larger goal of sustainable development.

Addressing climate change is paramount for achieving sustainable development goals. Climate action involves mitigating greenhouse gas emissions, adapting to climate impacts, and transitioning to a low-carbon economy. Tackling climate change contributes significantly to broader sustainable development objectives.Climate action aims to reduce carbon emissions, which not only mitigates climate change but also helps protect ecosystems, biodiversity, and natural resources. Preserving these resources is essential for sustainable development as they provide vital services such as clean air, water, and food security.Transitioning to renewable energy sources and implementing energy-efficient technologies creates new economic opportunities, spurring innovation and job creation. Investments in renewable energy infrastructure, sustainable agriculture, and green technologies can drive economic growth while reducing dependence on fossil fuels. Climate change disproportionately affects vulnerable communities, exacerbating social inequalities. Addressing climate change involves ensuring equitable access to resources, infrastructure, and opportunities for all. Policies aimed at climate resilience and adaptation can enhance the resilience of communities, particularly those most at risk from climate impacts.Climate change poses significant risks to public health, including heat-related illnesses, vector-borne diseases, and food insecurity. By reducing carbon emissions and promoting sustainable practices, climate action can improve air quality, enhance food security, and safeguard public health, contributing to overall well-being. Building resilience to climate change impacts is crucial for sustainable development. Investing in climate-resilient infrastructure, early warning systems, and disaster preparedness measures can help mitigate the impacts of extreme weather events and natural disasters, protecting lives and livelihoods. Climate change is a global challenge that requires collective action and cooperation among nations, businesses, and civil society. By fostering international collaboration and partnerships, climate action promotes peace, stability, and sustainable development worldwide.

C:Provide a detailed description of the key stakeholders affected by the problem.Consider the diverse perspectives involved in the issue

When we think about who's involved in dealing with climate change, it's important to realize that there are many different groups with their own interests and viewpoints. From governments crafting policies to businesses implementing sustainable practices, and from communities facing immediate impacts to future generations inheriting the consequences, a wide array of stakeholders play a role in shaping responses to climate change. Let's take a closer look at these groups and what they think about climate change.

Governments are essential to the development and execution of policies regarding adaptation and mitigation of climate change. They are in charge of establishing goals for reducing emissions, passing laws, offering financial incentives to switch to renewable energy sources, and funding efforts to build climate resilience. Decisions about policy can have a big impact on many different parts of society, like businesses, communities, and individuals.

NGOs, advocacy groups, community organizations, and grassroots movements play a vital role in raising awareness about climate change, advocating for policy change, and mobilizing public action. Since these groups are disproportionately impacted by the effects of climate change, they frequently speak for the interests of marginalized communities, indigenous peoples, and environmental justice activists.

Other than that, local communities, particularly those in vulnerable regions such as coastal areas, small island states, and arid regions, are directly impacted by climate change. Risks they must deal with include rising sea levels, severe weather, losing their means of subsistence, and being uprooted. Indigenous peoples, who often have deep ecological knowledge and cultural ties to their lands, are particularly affected by environmental changes and may face threats to their traditional way of life.

Lastly, young people are becoming more vocal in their demands for action on climate change since they will be the ones affected by the majority of current decisions made on environmental stewardship and climate policy. The effects of climate change will be borne by future generations, thus it is crucial to take their interests and viewpoints into account when making decisions.

Considering the perspectives and interests of these diverse stakeholders is essential for developing inclusive and effective strategies to address climate change and achieve sustainable development goals. Collaboration and dialogue among stakeholders with varying perspectives can lead to more comprehensive solutions that balance between environmental, social, and economic factors.

D: Consider the diverse interests involved in the issue
how your solution might benefit them.

<u>Individual</u>

The program to track and calculate carbon footprint offers individuals several benefits. Firstly, it helps them become more aware of how their daily actions impact the environment. With this knowledge, they can make smarter choices to reduce their carbon footprint and help reduce climate change. The program also allows users to set goals for themselves and track their progress over time, which can be motivating. Additionally, by making environmentally-friendly choices, individuals can often save money on things like energy bills. Plus, adopting habits like walking or biking instead of driving not only reduces emissions but also promotes a healthier lifestyle. Ultimately, engaging with the program connects individuals with a community of like-minded people who care about the planet, fostering a sense of shared responsibility and action.

Community

The program for tracking and calculating carbon footprint offers community groups various benefits. Firstly, it raises awareness about carbon emissions within the community, encouraging members to take action for a greener environment. This shared awareness fosters teamwork and a sense of responsibility for environmental stewardship. Moreover, the program facilitates collaboration among community members, empowering them to work together to find ways to lower emissions and promote sustainability. Through collaboration, community groups strengthen bonds and become more resilient to the impacts of climate change. Additionally, the program enables community groups to advocate for policies supporting environmental protection and educate others about the importance of sustainable living. Overall, by providing tools for tracking and reducing carbon footprints, the program helps create stronger and more sustainable communities.

Government

A program to track and calculate carbon footprints helps governments in many ways. It gives them useful information to make decisions about environmental rules and actions. They can focus their efforts where carbon emissions are highest. Also, the program helps them check if their plans to fight climate change are working. By involving people in tracking their carbon footprints, governments can spread awareness and support for climate action. Plus, the program helps enforce environmental rules by making sure everyone follows them. Overall, it lets governments take action to protect the environment and promote sustainability.

Education

Schools and educators play a crucial role in educating us about climate change and how to address it. By teaching students about climate science and sustainable living, they empower them to make a positive impact. Schools also offer hands-on learning experiences, like outdoor projects, that connect classroom lessons to real-world issues. Additionally, schools can set an example by adopting eco-friendly practices and using tools like carbon footprint calculators to help students understand their environmental impact. Ultimately, education helps us all become more mindful and take steps to protect our planet for the future.

d. Utilize the principles of Object-Oriented Programming to design a comprehensive solution that addresses the identified problem effectively. Consider the modularity, encapsulation, inheritance, and polymorphism concepts to create a robust and adaptable solution. Outline the specific features and functionalities of your proposed software solution.

To address the identified problem effectively with a comprehensive software solution, the principles of Object-Oriented Programming (OOP) will be employed to create a flexible, organized, and adaptable system. Below is a breakdown of the specific features and functionalities of the proposed software solution:

Modularity:

The software solution will be divided into modules, each responsible for specific tasks such as tracking and reducing carbon footprints, raising awareness, and facilitating collaboration among stakeholders.

Encapsulation:

Encapsulation will be employed to hide the internal implementation details of each module, exposing only the necessary interfaces for interaction. This ensures that each module can be modified or replaced without affecting the functionality of other modules.

Inheritance:

Inheritance can be utilized to create specialized modules or classes that inherit functionality from more general ones. For example, specific modules for tracking carbon footprints in different sectors (e.g., transportation, energy usage) can inherit common functionalities from a general carbon footprint tracking module.

Polymorphism:

Polymorphism allows different modules or classes to be treated interchangeably through a common interface. This enables flexibility in the software solution, allowing it to adapt to different requirements or scenarios.

Based on these principles, here are the specific features and functionalities of the proposed software solution:

User Authentication and Profiles:

Users can create accounts and profiles to access the software solution. Different user roles (individuals, communities, governments, educators) can have specialized functionalities and permissions tailored to their needs.

Carbon Footprint Tracking:

Users can input data about their daily activities, such as transportation methods, energy usage, and consumption habits. The software calculates users' carbon footprints based on the input data, providing insights into their environmental impact.

Goal Setting and Tracking:

Users can set personal or community goals for reducing carbon footprints. Progress towards these goals is tracked over time, motivating users to adopt more sustainable practices.

Educational Resources:

The software provides an information page that allows user access to educational resources on climate change, sustainable living, and environmental conservation. Schools and educators can utilize specialized materials and tools for teaching students about climate science and sustainability.

Data Analytics and Reporting:

The software collects and analyzes data on carbon footprints, community engagement, and policy impact. Users can generate reports and visualizations to track progress, identify trends, and inform decision-making.